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The Role of AI, Machine Learning, and Big Data in Digital Twinning: A Systematic Literature Review, Challenges, and Opportunities

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Abstract

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Digital twinning is one of the top ten technology trends in the last couple of years, due to its high applicability in the industrial sector. The integration of big data analytics and artificial intelligence/machine learning (AI-ML) techniques with digital twinning, further enriches its significance and research potential with new opportunities and unique challenges. To date, a number of scientific models have been designed and implemented related to this evolving topic. However, there is no systematic review of digital twinning, particularly focusing on the role of AI-ML and big

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addition to existing patents in the field. Also, we identified development-tools that can facilitate various levels of the digital twinning. Further, we designed a big data driven and AI-enriched reference architecture that leads developers to a complete DT-enabled system. Finally, we highlighted the research potential of AI-ML for digital twinning by unveiling challenges and current opportunities.

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